

Shubham Sanjay Patil

GenAI / AI Application Engineer

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PROFESSIONAL SUMMARY

GenAI / AI Application Engineer with 5+ years building production, AI-integrated software end to end. Specialized in LLM application engineering — Retrieval-Augmented Generation (RAG), LangGraph agents, LLM fine-tuning & quantization, and LLM observability / cost governance — on a strong full-stack base (Python/FastAPI, React, Go). At HPE, shipped 8+ enterprise GenAI services on the org-wide **One-AI** platform: a RAG knowledge assistant (ChromaDB + cross-encoder reranking + Azure OpenAI GPT-4), a LangGraph support agent, AI summarization & document-generation services, and a self-hosted Langfuse observability + LLM cost-attribution stack. Co-author of a technical paper on efficient LLM inference. OCI 2025 AI Foundations, KCNA, and Neo4j certified.

WORK EXPERIENCE

- Hewlett Packard Enterprise (HPE), Technical Solutions Consultant — AI Application Engineering** Nov 2022 – Present — Pune, India
- Engineered **One-AI**, a centralized enterprise AI platform delivering RAG knowledge assistants, an Azure OpenAI GPT-4 chat experience with enterprise guardrails, and developer/productivity tools behind a single secure gateway (Okta / Keycloak / JWT).
 - Built **HIPPO**, the platform's RAG pipeline — document ingestion & sync from 3 enterprise sources (Confluence, SharePoint, OneNote) into a **ChromaDB** vector store, cross-encoder (MiniLM) reranking, and grounded GPT-4 answers exposed as versioned FastAPI endpoints.
 - Developed **support-ai**, a **LangGraph** agent that blends internal knowledge retrieval with live web search to answer domain support questions.
 - Delivered **Asset Summary**, an Azure OpenAI summarization service with status/age-aware caching, scheduled pre-summarization (APScheduler), versioned prompt management and token accounting on async PostgreSQL; fed by a DuckDB→PostgreSQL aggregation pipeline on hourly Kubernetes CronJobs.
 - Stood up **AI Observability** on a self-hosted **Langfuse** stack and authored quarterly LLM token & cost-attribution reporting across projects — per-team / per-repo GenAI spend visibility (FinOps).
 - Developed **Z+ Application Security**, orchestrating **7 open-source security tools** (Nmap, Metasploit, OWASP ZAP, SQLMap, JMeter, OpenVAS, Dependency-Check) in parallel pipelines — Go (Gin) API + Python engine + React UI with RabbitMQ, MongoDB and Redis; containerized with Docker and shipped to Kubernetes via Jenkins / GitHub Actions.
- Infosys Ltd, Systems Engineer — Networking** Feb 2021 – Oct 2022 — Pune, India
- Built full-stack web applications for enterprise networking clients (Python/Flask, React, PostgreSQL, Docker/Kubernetes) on a microservices architecture.
 - Verizon – 5G Node Manager**: REST services monitoring 5G node health and network traffic with real-time alerting, parent-child node communication, historical logging and role-based access.
 - F5 Networks – BIG-IP**: automated integration & regression testing with the Robot Framework, increasing coverage and cutting manual test effort.

PROJECTS

- TuneX — LLM Fine-Tuning & Deployment Platform**
Hackathon Honorable Mention. Python (FastAPI), Celery, React/Vite, PostgreSQL/pgvector, Redis, Hugging Face.
- Fine-tunes open models for summarization, ticket classification, instruction-QA and email generation — with dataset pipelines, a model registry, quantization, and an inference loader/cache for serving fine-tuned models.
- ChatHUB — Embeddable AI Chatbot Platform**
FastAPI (LLM client), React, MySQL, and a publishable NPM widget package (Webpack).
- Drops a configurable, branded AI chatbot into any website with a single script include; users create projects and embed their own assistant.

SKILLS

GenAI & LLM: RAG · LLM agents (LangChain, LangGraph) · prompt engineering & management · fine-tuning (LoRA, INT8/INT4 quantization, distillation) · embeddings & vector search · cross-encoder reranking · Azure OpenAI (GPT-4) · Hugging Face · LLM observability & cost governance (Langfuse) · ChromaDB · pgvector · Neo4j

Backend: Python (FastAPI, Flask) · Go (Gin) · Node.js (Express) · ASP.NET Core · Java (Spring Boot) · REST · microservices · async · Celery · RabbitMQ · APScheduler · Keycloak · Okta · JWT · OAuth2

Frontend: React 18 · Redux Toolkit · TypeScript · Vite · Material-UI / Ant Design / Grommet · JavaScript · HTML · CSS

Data & Messaging: PostgreSQL · MySQL · MongoDB · MS SQL Server · Vertica · DuckDB · Redis · RabbitMQ

DevOps & Cloud: Docker · Docker Compose · Kubernetes · Jenkins · GitHub Actions · Nginx · Unicorn/Uvicorn · HashiCorp Vault · Oracle Cloud Infrastructure (OCI) · AWS

Practices: CI/CD · unit & integration testing (pytest) · Agile/Scrum (JIRA) · observability · Git / GitHub / GitLab

PUBLICATIONS

- V. Chaudhari, **S. Patil**, A. Chavan, "LLM Compression and Efficient Inference Platform for Enterprise AI Workloads," *HPE Parasparam 2026* — efficient LLM inference via quantization, distillation, dynamic batching, KV-cache optimization and workload-aware orchestration across hybrid-cloud/edge.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 **AI Foundations** Associate · OCI 2024 Foundations Associate · **KCNA** (Linux Foundation) · **Neo4j** Certified Professional · Neo4j Graph Data Science · AWS Fundamentals · Google Prompting Essentials (Modules 1–3)

EDUCATION

PG Dip. B.E.	C-DAC ACTS, Pune , Advanced Computing (PG-DAC) Government College of Engineering & Research (SPPU) , Mechanical Engineering First Class with Distinction — CGPA 8.29 / 10.	Aug 2022 June 2020
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